

Cover Letter

Revised submission to Wireless Personal Communications (R1 → R2)

To: The Editor-in-Chief, *Wireless Personal Communications* (Springer)

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Manuscript title: A Lightweight Complex-Valued CNN with Complex Squeeze-and-Excitation Attention for Single-Channel Blind Source Separation of Co-Frequency Communication Signals

Manuscript type: Original Research Paper

Revision: R1 → R2

Dear Editor,

We thank the editor and the reviewers for the thorough and constructive critique of our original submission. The R1 review identified genuine weaknesses in the manuscript — the missing multipath in Eq. (1), the under-tested “co-frequency” claim, the soft baseline set, the insufficient statistical reporting, and several overclaims — and we have addressed every one of them in this revision. A detailed point-by-point response is provided in `reviewer_response.md` (27 items, R1.1–R1.12, R2.1–R2.10, R3.1–R3.3). Below we summarise the substantive changes for the editor’s convenience.

Major changes since R1.

1. **Expanded baselines.** The R1 manuscript compared only against the no-SE ablation, a real-valued baseline, and a complex Conv-TasNet. In R2 we add (i) two *parameter-matched* complex-CNN baselines (no-SE at $H=70$ giving 242K params, real-valued at $H=80$, $L=12$ giving 237K params) to disentangle the contribution of the SE block from the parameter budget, and (ii) two *cross-domain* communication-signal baselines (CNSE by Hou & Gao 2022 and S4-UNET by Gao et al. 2026) re-implemented from the primary sources and scaled down to fit the 8 GB GPU used in our setting.
2. **Statistical reporting.** Every evaluable configuration is now reported over **five** random seeds (42–46) with sample standard deviation. We add a dedicated subsection (Section 4.8) reporting paired Wilcoxon signed-rank tests between C-SE and every baseline on per-seed SDR. We are explicit that *no comparison reaches statistical significance at $\alpha = 0.05$* — the smallest attainable two-sided p at $N=5$ pairs is 0.0625 — and reposition the paper’s narrative around per-parameter efficiency and per-seed reproducibility rather than raw SDR dominance.
3. **Equation (1) and scope.** Eq. (1) now includes the multipath channel h_k with explicit 3-tap complex Gaussian definition, normalisation, and unit-energy constraint. A new paragraph after the equation states the scope: the generator uses symbol-aligned mixtures with per-source carrier offset within ± 5 Hz; relaxing symbol alignment is left to future work.
4. **Generalisation to the co-frequency limit.** We trained two new models (proposed C-SE and matched no-SE) with the carrier-frequency gap drawn from $U(0, 5)$ Hz per sample. Evaluating at gaps from 0 to 500 Hz (new Section 4.7, Fig. `micro_freq`) shows the proposed C-SE holds $\text{SDR} \approx 3.3$ dB across the entire range, including the previously unseen co-frequency limit $\Delta f = 0$ Hz ($\text{SDR} = 3.33$ dB). The matched no-SE baseline

trained under the same schedule still collapses to $\text{SIR} \approx 21$ dB at every gap, confirming the gap is structural.

5. **Honest positioning.** “First” is now softened throughout to “to the best of our knowledge, the first phase-preserving, real-gated complex-valued SE design specifically evaluated for lightweight single-channel blind separation of near-co-frequency communication signals.” “SOTA” is removed from the Conv-TasNet label. “Is in fact optimal” is replaced with “is sufficient” for the scale-mode ablation. The Discussion section acknowledges prior complex-domain work (CSENet 2022, complex-valued SE 2024/2025) and identifies the C-SE’s specific contribution as the phase-preserving real-gated design.

Headline numbers (5 seeds, SNR = 10 dB). The proposed C-SE reaches $\text{SDR} = 2.31(62)$ dB at 235 K parameters (5 seeds). The matched no-SE complex CNN reaches 1.81(48) dB (5 seeds); the matched real-valued CNN reaches 2.20(59) dB (5 seeds); the Complex Conv-TasNet reaches 3.23(40) dB (3 seeds, 817K params). CNSE (6.69M) reaches 2.38(91) dB (3 seeds) and S4-UNET (1.57M) reaches 3.05 dB (3 seeds). C-SE is therefore competitive at the 235 K-parameter scale on the dimensions of partial collapse-resistance (2 of 5 vs. 4 of 5 for matched no-SE), per-parameter efficiency, and predictable runtime; we make no claim of statistical dominance.

Reproducibility and ethics.

- The manuscript reports per-seed numbers for every evaluated configuration; the full implementation will be released as open-source code (MIT) upon acceptance.
- `docs/EXPERIMENT_LOG.md` at the project root provides a complete reproducibility record with exact commands, hyperparameters, and per-seed SDR/SI-SDR/SIR values for all 7 experimental phases and 44 training runs.
- `paper1_cnn_se/data_radioml.py` ships an optional loader for the public RadioML 2016.10A corpus (not used in headline results; left for the community to extend the evaluation to over-the-air and public I/Q data).
- The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal.
- The authors have no conflicts of interest to declare.
- No human subjects or animal experiments are involved.

Author contributions. B. Nong conceived the C-SE design, ran the multi-seed experiments, and drafted the manuscript. W. Fu contributed the matched-baseline analysis and the scale-mode / pooling ablations. Z. Jiang contributed the cross-domain baseline re-implementations (CNSE, S4-UNET) and the statistical-significance analysis. All authors reviewed and approved the final manuscript.

Funding. This work received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Suggested reviewers. We have no specific reviewer suggestions; we trust the editorial board to identify expert reviewers in signal processing for wireless communications.

Thank you for considering our revised manuscript. We respectfully request a re-evaluation.

Sincerely,

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